

ASTRALINK — AI-BASED VIRTUAL CAMERA TRACKING SYSTEM FOR MOBILE FSOC

Coarse Alignment - Performance Report

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1. RUN CONFIGURATION

Frames Processed

67

Duration

2.01 s

Processing FPS

148.7

Avg Processing

6.72 ms

Max Processing

8.46 ms

2. TRACKING PERFORMANCE

Acquisition Time

0.60 s

<= 2 s

Average Tracking Error

1.94 px

Maximum Tracking Error

4.00 px

<= 10 px

RMSE Tracking Error

2.13 px

Centroid Error Avg/Max

0.24 / 0.50 px

ground truth only

Lock Retention

100.00 %

Target Loss

0.00 %

< 5 %

Re-acquisition Avg/Max

—

<= 1 s

Successful Re-acq.

0

0 loss events

3. PS REQUIREMENT CHECK

Parameter	Measured	Requirement	Status
Acquisition Time	0.60	<= 2.0 s	PASS
Tracking Error (Max)	4.00	<= 10 px	PASS
Center Error (Final)	4.00	<= 10 px	PASS
Target Loss	0.00 %	< 5%	PASS
Re-acquisition Time	—	<= 1.0 s	PENDING
Processing Speed	148.7 FPS	>= 20 FPS	PASS

Ground truth was available: centroid accuracy is measured.

OVERALL: PENDING

Source: live simulation | Frames: 67 | FPS: 148.7